Week 1 Internship Tasks

Submitted By : Meenakshi Muthukumaran

Organization : InterNova

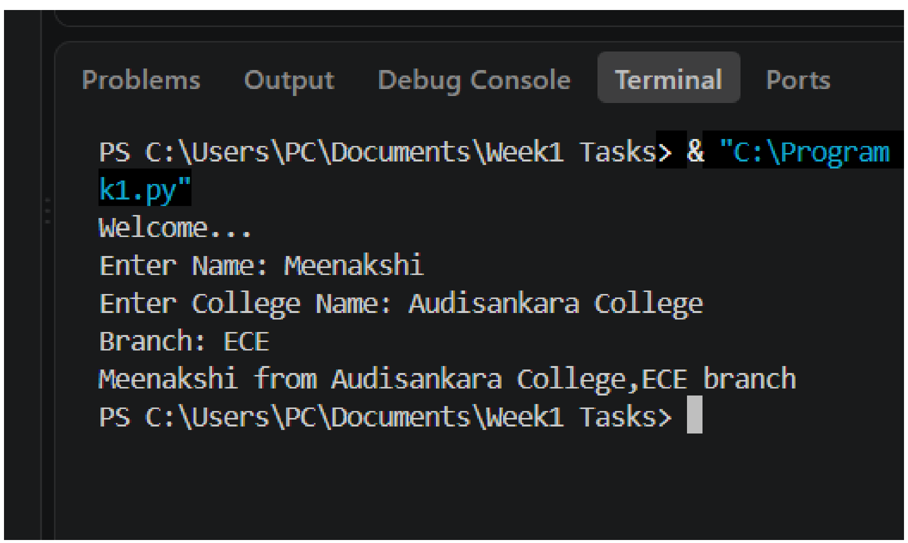
Domain : Data Analytics

Task 1

Source Code:

```
print("Welcome...")
name=input("Enter Name: ")
collegeName=input("Enter College Name: ")
branch=input("Branch: ")
print(f"{name} from {collegeName},{branch} branch")
```

Output:



Task 1

Brief Explanation:

1)The program begins by displaying a welcome message using the print() function.

2)It then prompts the user to enter their **name** using the input() function and stores it in the variable name.

3)Next, it asks the user to enter their **college name** and stores it in the variable collegeName.

4)The program then accepts the user's **branch** and stores it in the variable branch.

5)After collecting all the required information, it uses an **f-string** to combine the entered values into a single formatted sentence.

6)Finally, the formatted output is displayed on the screen using the print() function.

Task 2

Source Code:

```
pincode=123567
city="Hyderabad"
temperature=36.9
is_raining=False
#Displaying variable and their data type

print("Pincode:",pincode)
print("Data Type:",type(pincode))

print("City: ",city)
print("Data Type:",type(city))

print("Temperature: ",temperature)
print("Data Type:",type(temperature))

print("Is_Raining:",is_raining)
print("Data Type:",type(is_raining))
```

Task 2

OUTPUT:

```
Problems    Output    Debug Console    Terminal    Ports

PS C:\Users\PC\Documents\Week1 Tasks> & "C:\Program 
k2.py"
Pincode: 123567
Data Type: <class 'int'>
City:  Hyderabad
Data Type: <class 'str'>
Temperature:  36.9
Data Type: <class 'float'>
Is_Raining: False
Data Type: <class 'bool'>
PS C:\Users\PC\Documents\Week1 Tasks> 
```

Brief Explanation:

The program declares variables of different data types: integer, float, string, and boolean.Each variable is assigned a value.The print() function is used to display the value of each variable.The type() function is used to identify and display the data type of each variable.

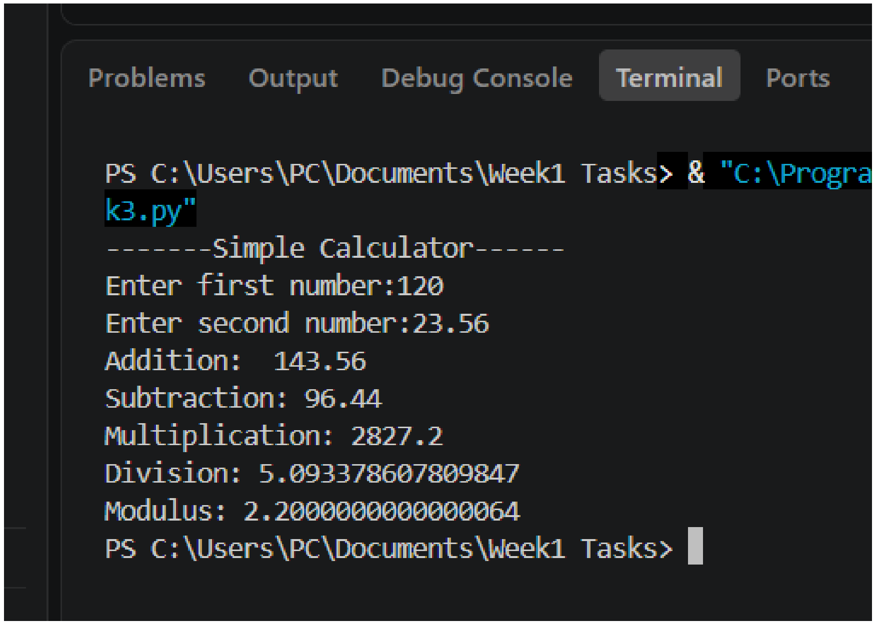
Task 3

Source Code:

```
print("-------Simple Calculator------")
num1=float(input("Enter first number:"))
num2=float(input("Enter second number:"))

print("Addition: ",num1+num2)
print("Subtraction:",num1-num2)
print("Multiplication:",num1*num2)
if(num2!=0):
   print("Division:",num1/num2)
   print("Modulus:",num1%num2)

else:
   print("Cannot perform division operation")
   print("Cannot perform modulus operation")
```

Task 3

OUTPUT:



Brief Explanation:

The program accepts two numbers as input from the user.It performs the basic arithmetic operations: addition, subtraction, multiplication, division, and modulus.Each operation is calculated using the corresponding Python operator (+, -, *, /, %).The results of all the operations are displayed on the screen using the print() function.

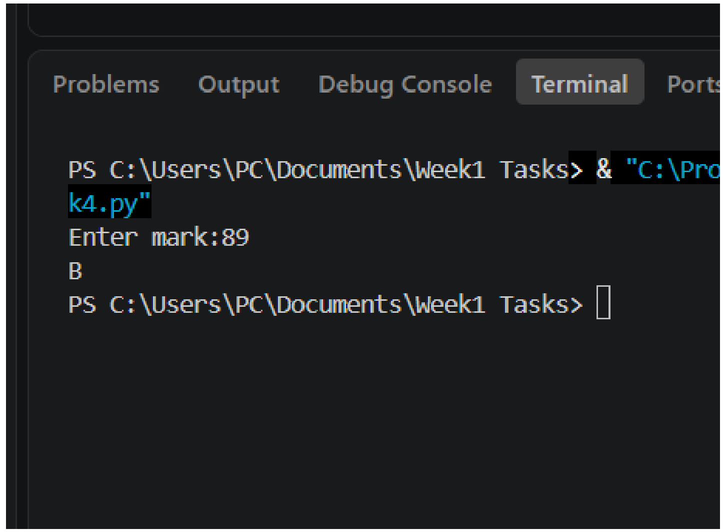
Task 4

Source Code:

```
mark=float(input("Enter mark:"))

if(mark<0 or mark>100):
  print("Invalid Marks! please enter mrks between 0 and 100")
else:
  if(mark>=90):
    print("A")
  elif(75<=mark<=89):
    print("B")
  elif(60<=mark<=74):
    print("C")
  else:
    print("fail")
```

Task 4

OUTPUT:



Brief Explanation:

The program accepts the student's marks as input from the user.It uses if, elif, and else statements to check the range of marks.Based on the entered marks, the program assigns the appropriate grade:

90 and above → Grade A
75 to 89 → Grade B
60 to 74 → Grade C
Below 60 → Fail

Finally, the corresponding grade is displayed on the screen.

Task 5

Source Code:

```
#print numbers from 1 to 20 using for loop
for i in range(1,11):
    print(i)

# print the multiplicaton table of any number
num=int(input("Enter a integer: "))
for i in range(1,11):
    print(f"{num} * {i} = {num * i}")

#print even number from 1 to 50 uing a while loop
for i in range(1,51):
    if i%2==0:
        print(i)
```

Task 5

OUTPUT:

```
PS C:\Users\PC\Documents\Week1
k5.py"
1
2
3
4
5
6
7
8
9
10
```

```
Problems   Output   Debug Console   Terminal   Ports

6 * 1 = 6
6 * 2 = 12
6 * 3 = 18
6 * 4 = 24
6 * 5 = 30
6 * 6 = 36
6 * 7 = 42
6 * 8 = 48
6 * 9 = 54
6 * 10 = 60
Even number over the range of 1-50 are:
2 4 6 8 10 12 14 16 18 20 22 24 26 28 30 32 34 36 38 40 42 44 46 48 50
PS C:\Users\PC\Documents\Week1 Tasks>
```

Brief Explanation:

The program prints numbers from **1 to 10** using a for loop.It accepts an integer from the user and displays its multiplication table from **1 to 10**.It then prints all **even numbers from 1 to 50** by checking whether each number is divisible by **2**.

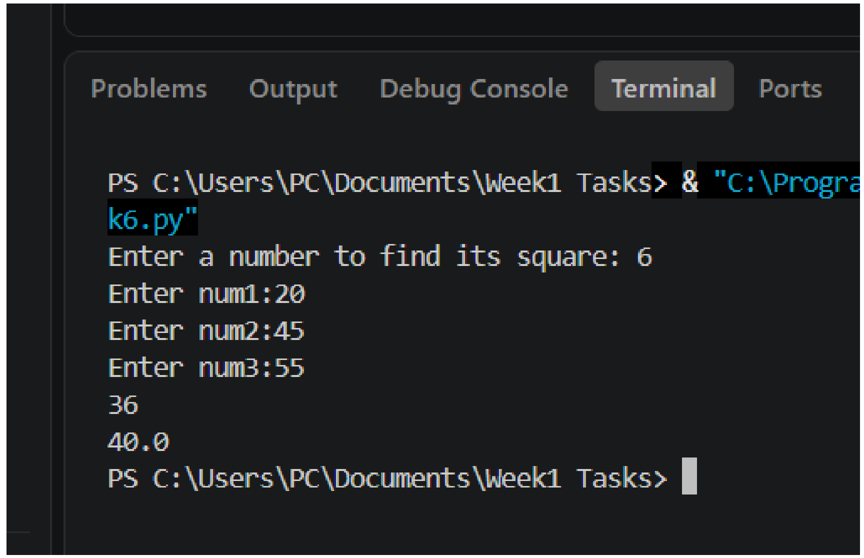
Task 6

Source Code:

```
def square(val):
    result=val*val
    return result

def average(num1,num2,num3):
    avg=(num1+num2+num3)/3
    return avg

val=int(input("Enter a number to find its square: "))
num1=int(input("Enter num1:"))
num2=int(input("Enter num2:"))
num3=int(input("Enter num3:"))

print(square(val))
print(round(average(num1,num2,num3),2))
```

Task 6

OUTPUT:



Brief Explanation:

The program defines two user-defined functions: one to calculate the **square of a number** and another to calculate the **average of three numbers**.It accepts the required input values from the user.The functions are called with the user input to perform the calculations.Finally, the program displays the square and the average of the entered numbers.

Task 7

Source Code:

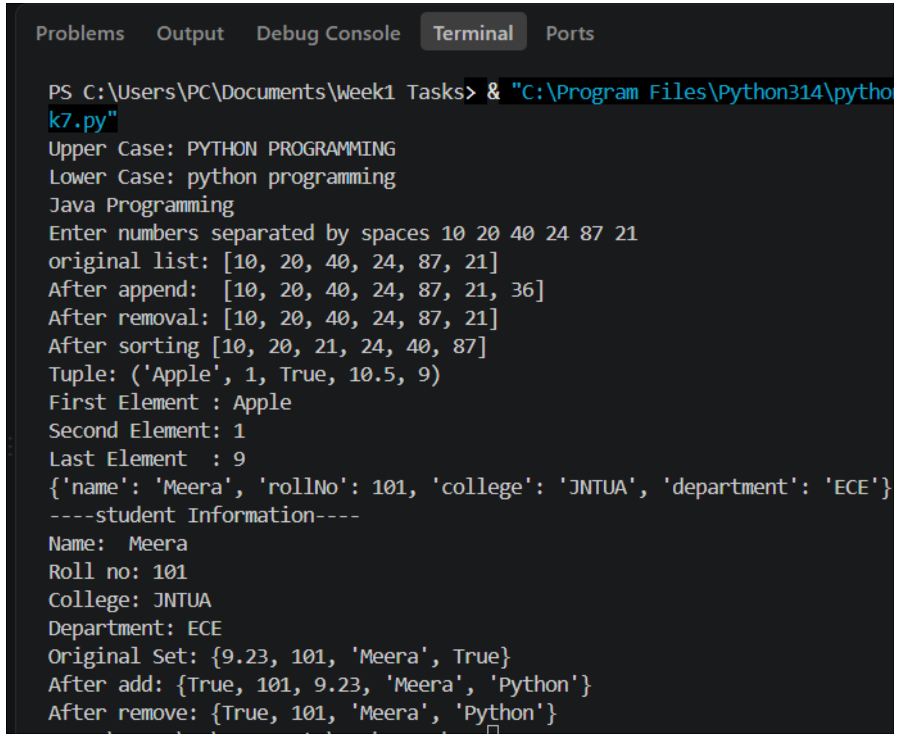
```
#String operations :upper(), lower(), replace(), find()
word="Python Programming"

print("Upper Case:",word.upper())
print("Lower Case:",word.lower())

old_word="Python"
new_word="Java"
print(word.replace(old_word,new_word))
search=word.find('gram')

#List operations (append(), remove(), sort())
lst=list(map(int,input("Enter numbers separated by spaces ").split()))
print("original list:",lst)
lst.append(36)
print("After append: ",lst)
lst.remove(36)
print("After removal:",lst)
lst.sort()
print("After sorting",lst)

# Tuple Creation and Indexing
tup = ("Apple",1, True, 10.5, 9)
print("Tuple:", tup)
print("First Element :", tup[0])
print("Second Element:", tup[1])
print("Last Element  :", tup[-1])
```

Task 7

```
#Dictionary storing student information

student={"name":"Meera",
    "rollNo":101,
    "college":"JNTUA",
    "department" :"ECE"}
print(student)
print("----student Information----")
print("Name: ",student["name"])
print("Roll no:",student["rollNo"])
print("College:",student["college"])
print("Department:",student["department"])
#Set operations (add(), remove())
data = {101, "Meera", 9.23, True,1}
print("Original Set:", data)
data.add("Python")
print("After add:", data)
data.remove(9.23)
print("After remove:", data)
```

Task 7

OUTPUT:




```
PS C:\Users\PC\Documents\Week1 Tasks> & "C:\Program Files\Python314\pytho
k7.py"
Upper Case: PYTHON PROGRAMMING
Lower Case: python programming
Java Programming
Enter numbers separated by spaces 10 20 40 24 87 21
original list: [10, 20, 40, 24, 87, 21]
After append:  [10, 20, 40, 24, 87, 21, 36]
After removal: [10, 20, 40, 24, 87, 21]
After sorting [10, 20, 21, 24, 40, 87]
Tuple: ('Apple', 1, True, 10.5, 9)
First Element : Apple
Second Element: 1
Last Element  : 9
{'name': 'Meera', 'rollNo': 101, 'college': 'JNTUA', 'department': 'ECE'}
----student Information----
Name:  Meera
Roll no: 101
College: JNTUA
Department: ECE
Original Set: {9.23, 101, 'Meera', True}
After add: {True, 101, 9.23, 'Meera', 'Python'}
After remove: {True, 101, 'Meera', 'Python'}
```

Brief Explanation:

The program demonstrates basic operations on Python data structures.It performs string operations using upper(), lower(), replace(), and find().It demonstrates list operations such as append(), remove(), and sort().It creates a tuple and accesses its elements using indexing.It stores and displays student information using a dictionary.It performs set operations by adding and removing elements.

Task 8

SOURCE CODE:

```
with open("introduction.txt", "w") as file:
    file.write("My name is Meenakshi Muthukumaran.\n")
    file.write("I am interested in Python programming and Data 
Analytics.\n")
    file.write("I enjoy learning new concepts and working on real-world 
projects.\n")
    file.write("I am eager to develop my skills and grow as a 
professional.")

file.close()

# Reading the file
file = open("introduction.txt", "r")
content = file.read()
print("File Contents:\n")
print(content)
file.close()
```

Task 8

OUTPUT:

task1.py task2.py task3.py task4.py task5.py task6.py task7.py

introduction.txt

```
My name is Meenakshi Muthukumaran.
I am interested in Python programming and Data Analytics.
I enjoy learning new concepts and working on real-world projects.
I am eager to develop my skills and grow as a professional.
```

Brief Explanation:

The program creates a text file and writes the user's introduction into it.It then opens the same file in read mode.The contents of the file are read and displayed on the screen.

Task 9

SOURCE CODE:

```
# Mini Project
# Student Management System

students=[]
def add_student():
  name=input("Enter Name: ")
  rollNo=input("Enter roll no:")
  age=input("Enter age:")
  college=input("Enter college name:")
  branch=input("Enter branch: ")
  data={"Name":name,
      "rollNo":rollNo,
      "Age":age,
      "College":college,
      "Branch":branch
   }
  students.append(data)
  print("Added student info successfully")
```

Task 9

```
def display_info():
   if(len(students)==0):
        print("No record found")

   else:
       print("------Students Records----- ")
       for student in students:
         print("name :",student["Name"])
         print("roll no:",student["rollNo"])
         print("Age: ",student["Age"])
         print("College: ",student["College"])
         print("branch:",student["Branch"])
         print("------------")
```

Task 9

```
def searchStudent():
  std_name=input("Enter student name to fetch details: ")
  for student in students:
    if std_name.lower()==student["Name"].lower():
      print("Record found!")
      print("Name:",student["Name"])
      print("roll no:",student["rollNo"])
      print("Age: ",student["Age"])
      print("College: ",student["College"])
      print("branch:",student["Branch"])
      return
    else:
      print("Record not found")
def delete_student():
  name=input("Enter student name to delete:")
  for student in students:
    if student["Name"].lower()==name.lower():
      students.remove(student)
```

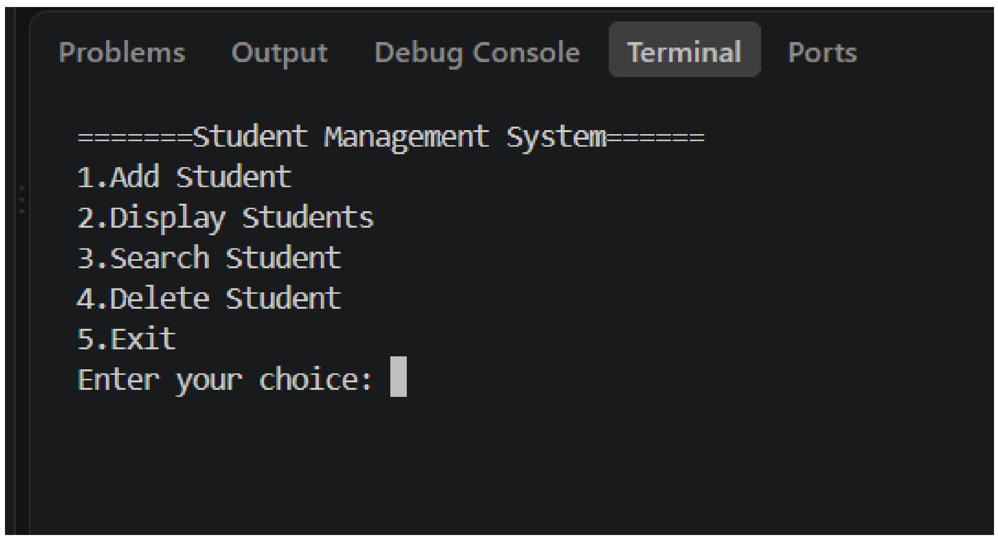
Task 9

```
print("Removed data")
      return
  else:
    print("Student not found")
while True:
  print("=======Student Management System======")
  print("1.Add Student")
  print("2.Display Students")
  print("3.Search Student")
  print("4.Delete Student")
  print("5.Exit")
  choice=int(input("Enter your choice: "))
  if choice==1:
   add_student()
  elif choice==2:
   display_info()
  elif choice==3:
   searchStudent()
  elif choice==4:
   delete_student()
```

Task 9

```
elif choice==5:
    print("Thank you")
    break
  else:
    print("Invalid choice.Please try again")
```

Output:



Task 9

Brief Explanation:

The program implements a simple **Student Record Management System**.
It allows the user to **add**, **display**, **search**, and **delete** student records through a menu-driven interface.
Student information is stored using a **list of dictionaries**.
Functions are used to organize the program and perform each operation efficiently.

Concepts Used:
Functions
Lists
Dictionaries
if-elif-else
Loops
User input
Menu-driven programming